# 獸醫臨床及影像診斷學

Digestive system

# Oral exam should be performed as part of each physical exam



**Adults:**
**teeth, excessive tartar,**
**tooth fracture,**
**oral masses**

**Puppies:**
**occlusion,**
**retained deciduous teeth**

# Dental tartar and root abscessation



# Gingival mass (epulis)



Gingival mass (epulis) in a Doberman Pinscher.

# Examine for evidence of anemia, icterus or petechia



Pale yellow mucous membranes in a dog with hemolytic anemia.

# Petechia-
# Immune mediated thromobopenia


Oral petechiae and pallor in a dog with immune mediated thrombocytopenia.

# 犬貓齒式

表 11.2　犬貓的上顎及下顎，每四分之一部分的牙齒組成

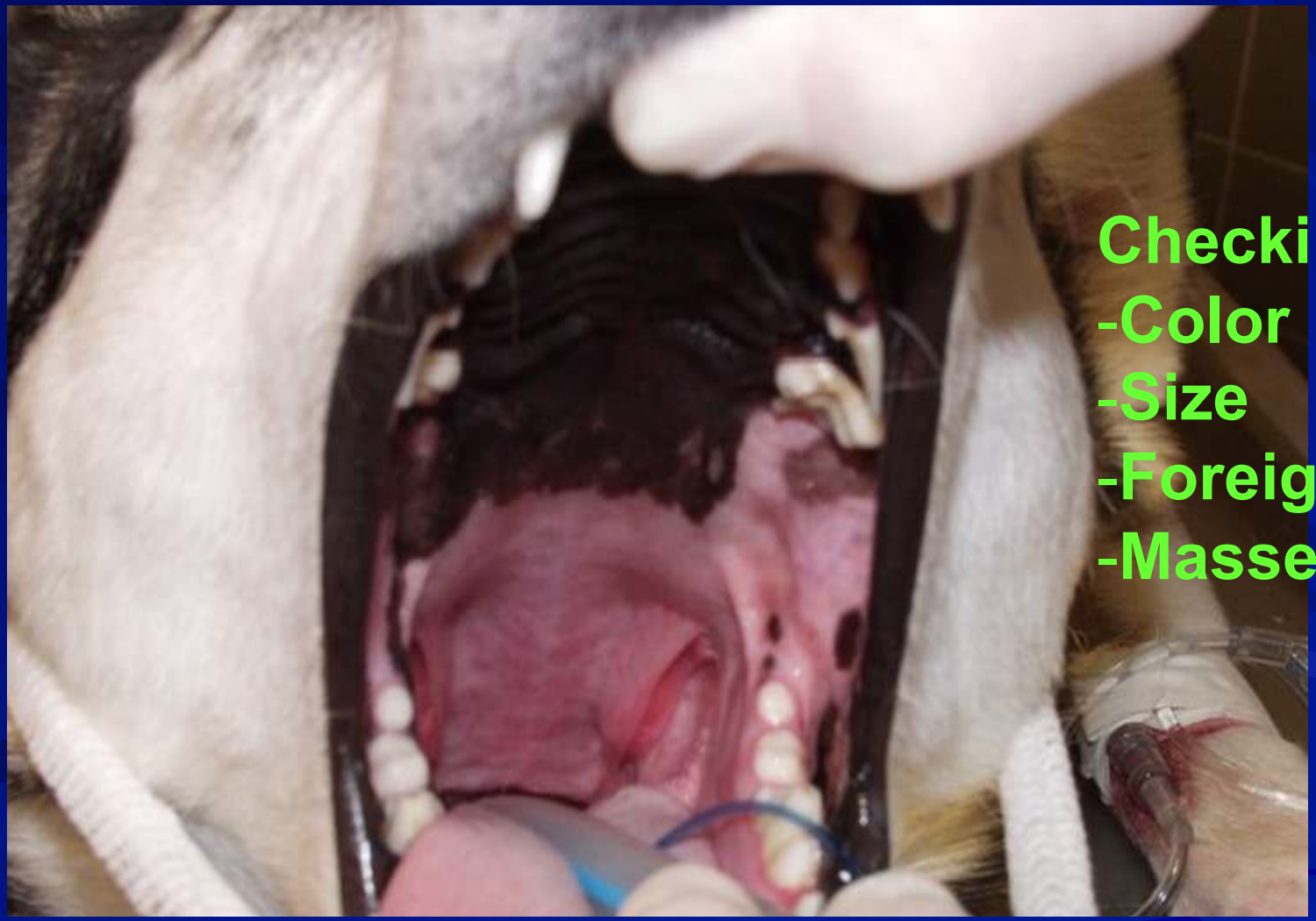
| | 犬 | 貓 |
|---|---|---|
| 乳齒（上顎） | 3i, 1c, 3p | 3i, 1c, 3p |
| （下顎） | 3i, 1c, 3p | 3i, 1c, 2p |
| 永久齒（上顎） | 3I, 1C, 4P, 2M | 3I, 1C, 3P, 1M |
| （下顎） | 3I, 1C, 4P, 3M | 3I, 1C, 2P, 1M |

I＝門齒，C＝犬齒，P＝前臼齒，M＝臼齒（乳齒以小寫字母標記）

# Retained deciduous teeth



# Tonsil examination



Checking for
-Color
-Size
-Foreign bodies
-Masses

- Probe the tonsilar crypt
- Palpate the hard palate
- Examine the sublingual salivary gland

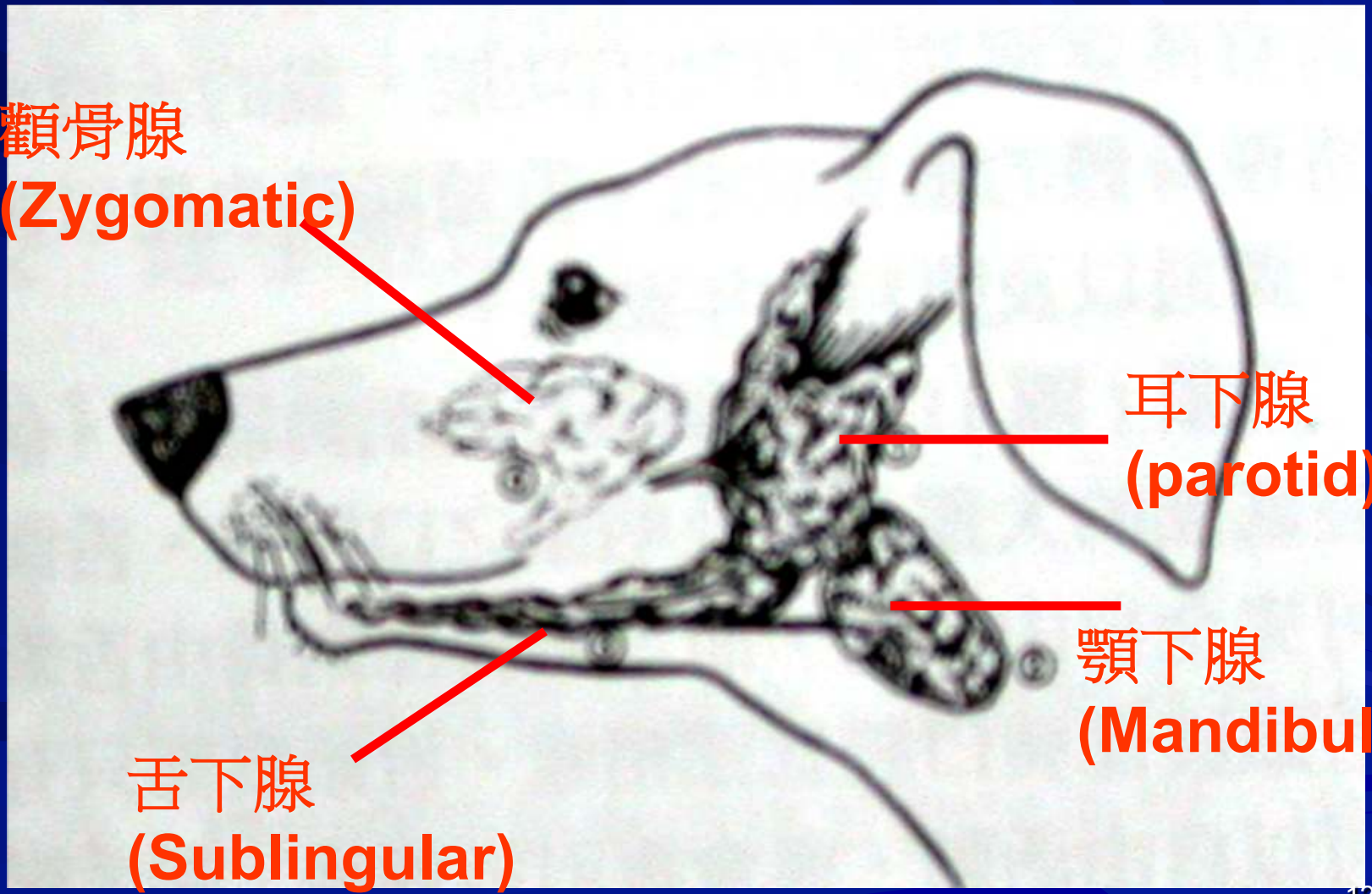

Probing the tonsilar crypt.

# Palpating the hard palate

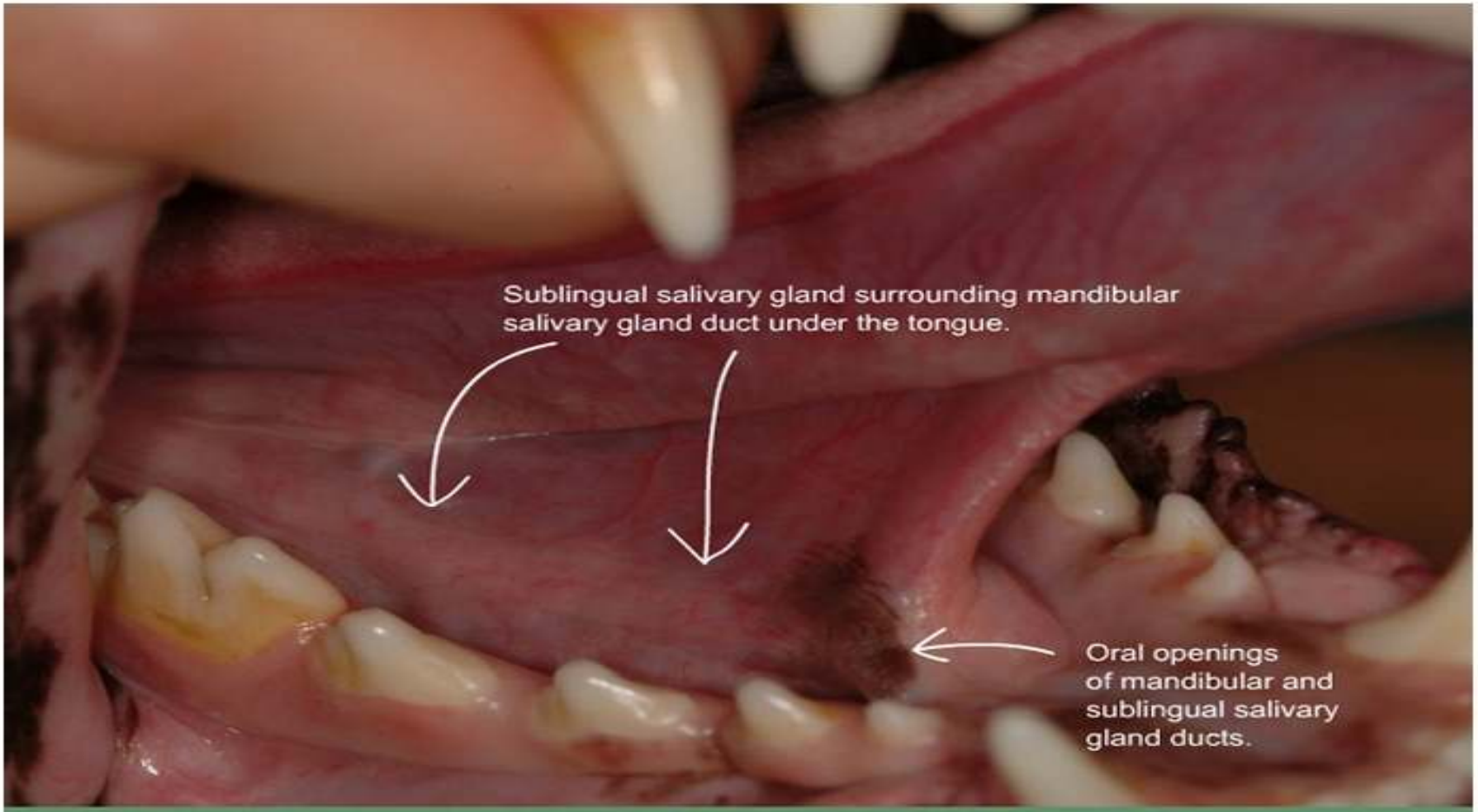


Palpating the hard palate.

# Salivary glands

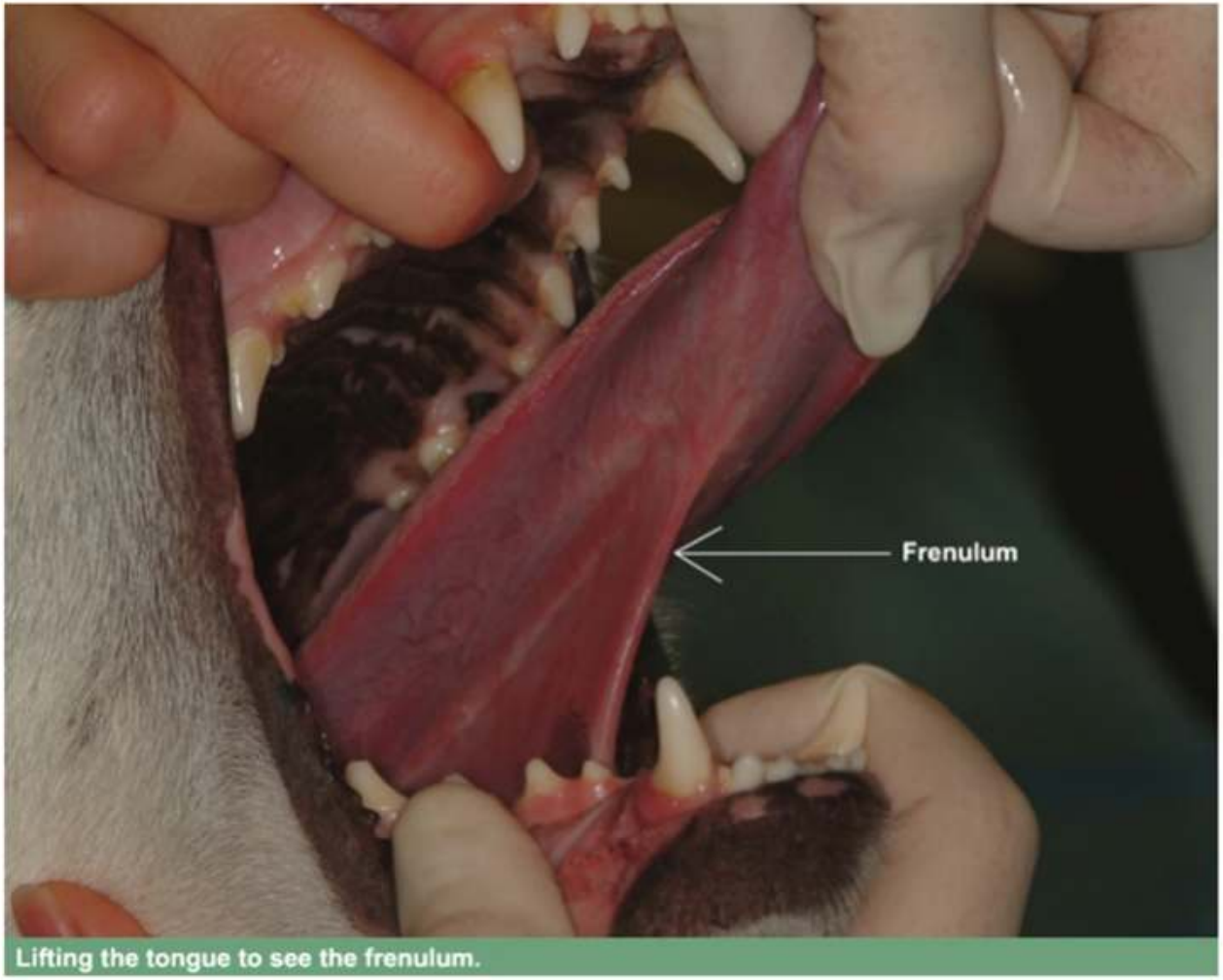


# Sublingual salivary gland




The sublingual salivary glands are normally barely visible in the intermandibular space under the tongue.

# Frenulum




Lifting the tongue to see the frenulum.

# Tongue ulcer

- Systemic lupus erythematous with polyarthritis




Ulcers on the edge of the tongue in a 7 year old German Shepherd with polyarthritis.
This dog had systemic lupus erythematosus.

11yr female dachshund
1 months history of sneezing after eating or drinking



# Oral checking procedure



# Cat’s tongue



The cat's tongue, used for grooming is covered with firm spines (papillae).

# Tongue ulcer of a cat
# - Calici virus infection


This cat has ulcers on its tongue caused by Calici virus infection.

1yr cat
3 days history of vomiting
3 days history of unwillingness to eat
7-10% Dehydration



String foreign body

# String foreign body of a cat



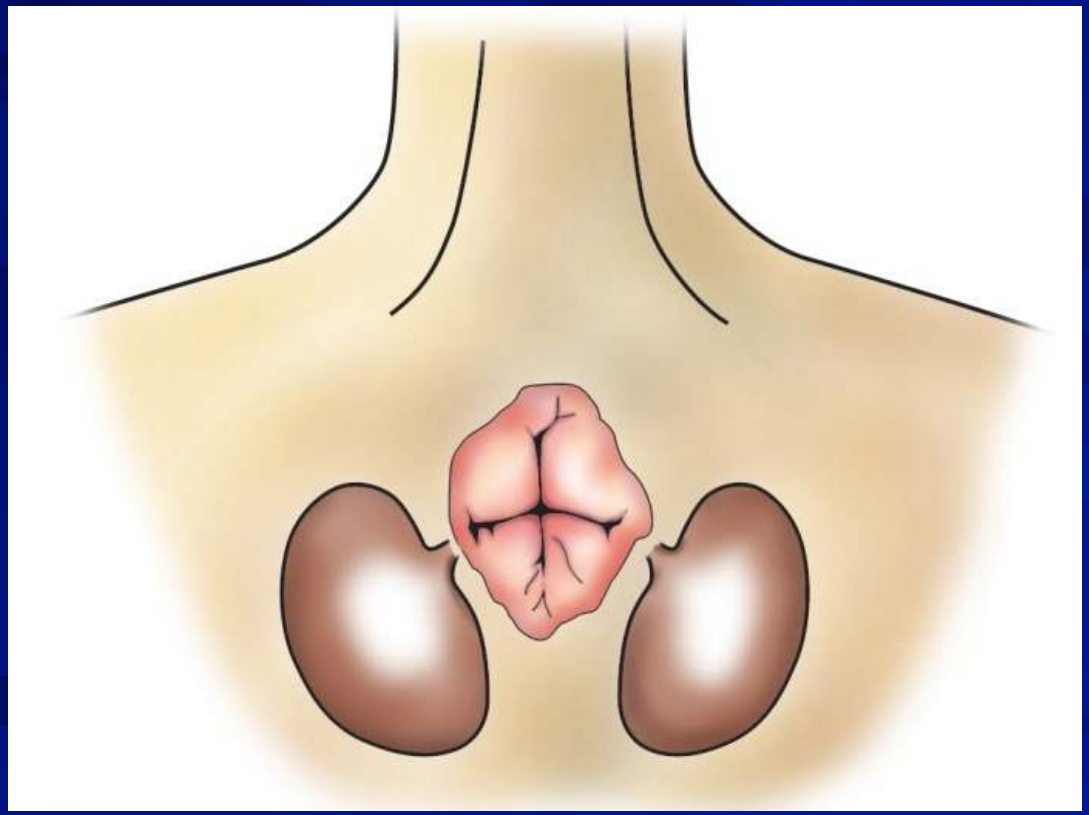
# Examining under the tongue of a cat



# Anal sac palpation and expression



# Anatomy of the anal sac



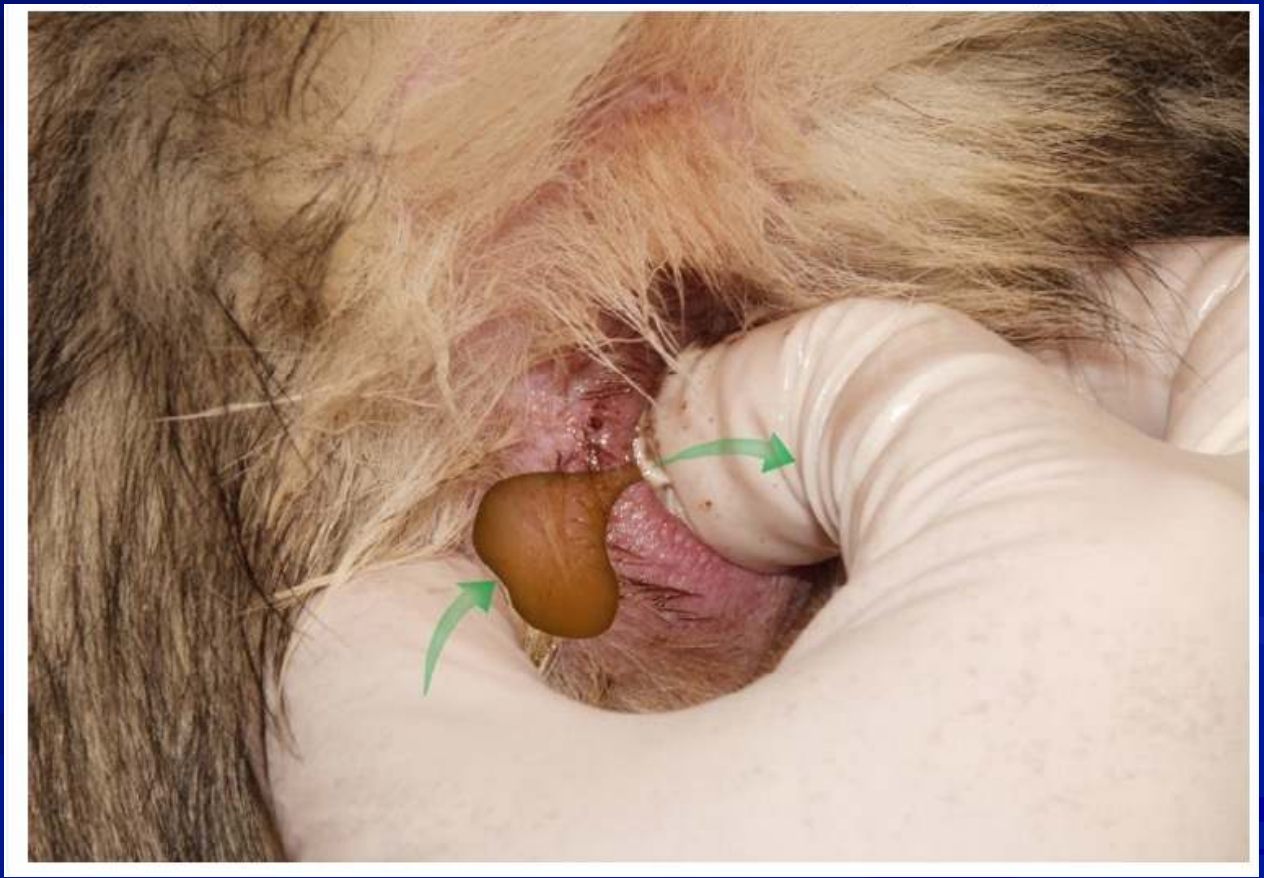
5 and 7 o’clock
(4 and 8 o’clock)
related to the anus

# Procedure of anal sac palpation and expression



- Insert finger into rectum to palpate the region
- Identify the anal sac between index finger and thumb

# Squeeze the anal sac from the ventral surface towards the anal sac opening



# Once emptied, palpate the empty anal sac for any thickenings or masses





Normal anal sac contents can vary in color and consistency.
Most often the secretions are yellow, grey or brown in colour.

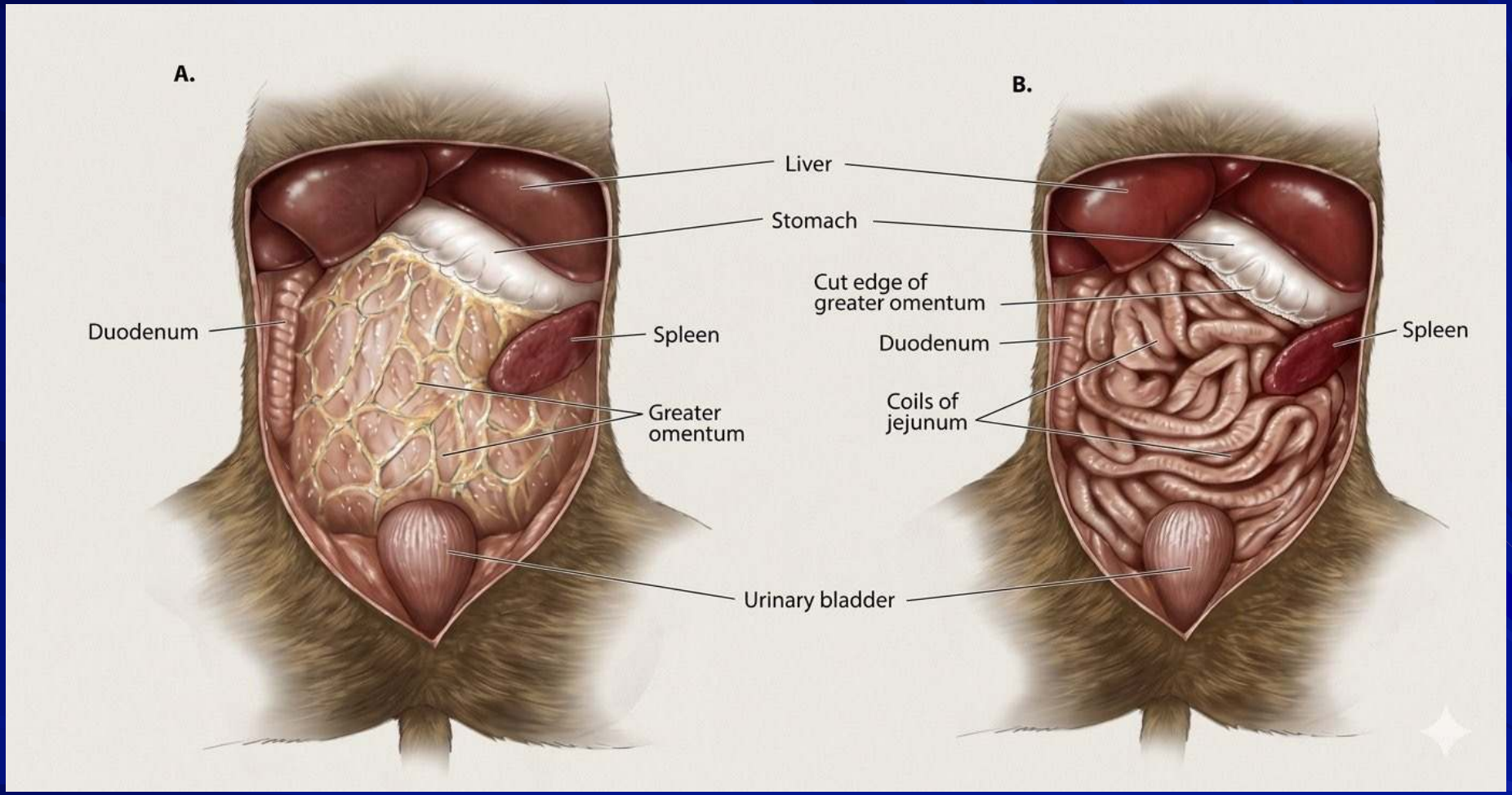
# 嘔吐與食物逆流

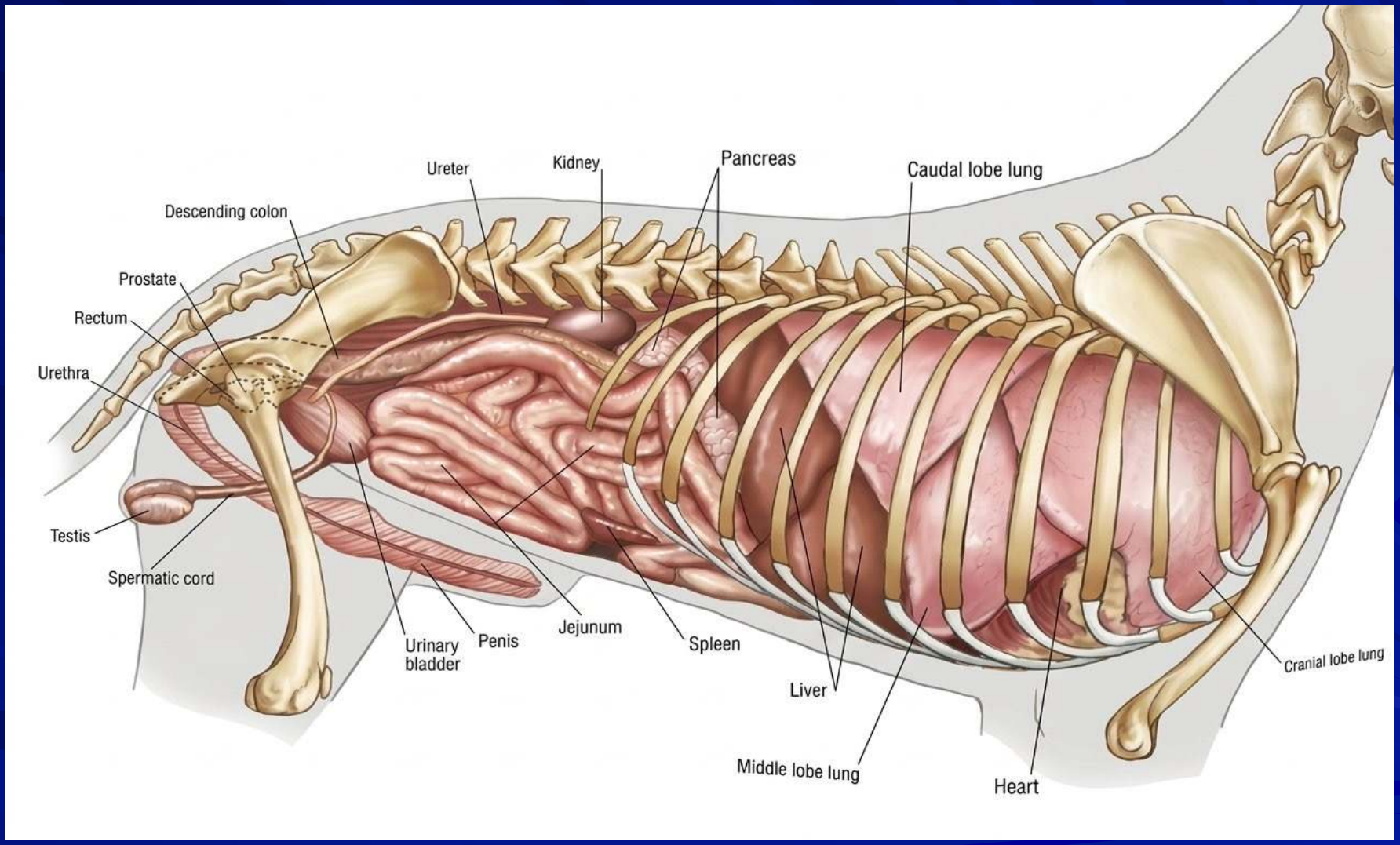
- 嘔吐
  - 腹壁肌肉主動收縮
  - 神經性刺激腦幹的嘔吐中樞
- 逆流
  - 咽部痲痺
  - 食道括約肌鬆弛不能
  - 食道麻痺
  - 幽門狹窄

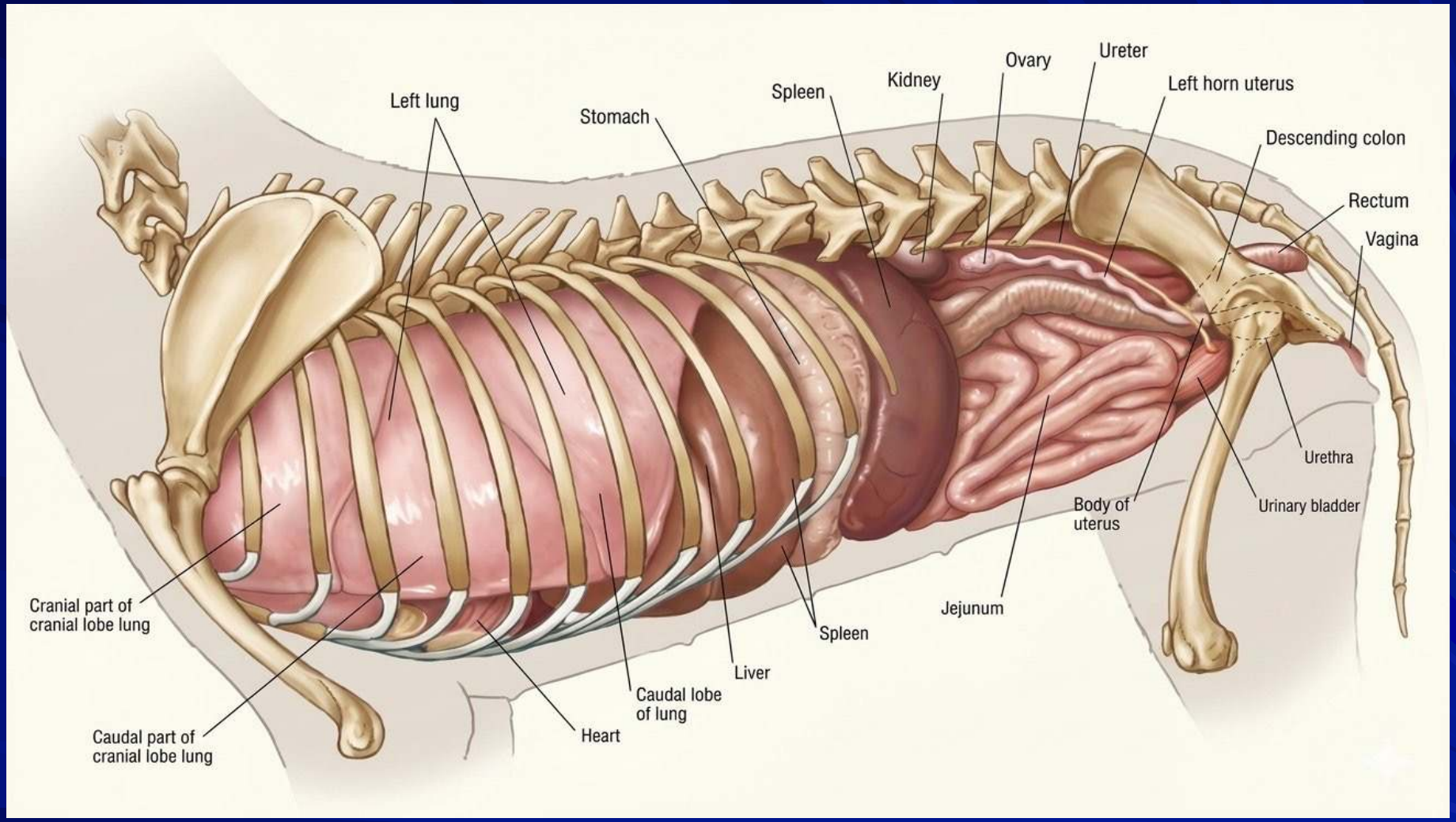
# 小腸與大腸性下痢

- 小腸性下痢：病毒感染使小腸絨毛萎縮、小腸球蟲寄生等
- 大腸性下痢：腐敗食物、過敏、纖維多寡等等

| | 小腸性下痢 | 大腸性下痢 |
|---|---|---|
| 排便頻率 | 低 | 高 |
| 裡急後重 | 罕見 | 常見(持續) |
| 糞便的體積 | 大 | 小 |
| 糞便含黏液 | 罕見 | 常見 |
| 糞便含血液 | 罕見 | 常見 |
| 劇渴 | 常見 | 罕見 |
| 多食 | 常見 | 罕見 |
| 腹鳴與脹氣 (Borborygmi and flatulence) | 常見 | 罕見 |
| 體重下降 | 常見 | 罕見 |


A.
B.
Liver
Stomach
Duodenum
Spleen
Greater omentum
Urinary bladder
Cut edge of greater omentum
Duodenum
Spleen
Coils of jejunum


Ureter
Kidney
Pancreas
Caudal lobe lung
Descending colon
Prostate
Rectum
Urethra
Testis
Spermatic cord
Urinary bladder
Penis
Jejunum
Spleen
Liver
Middle lobe lung
Heart
Cranial lobe lung


Left lung
Stomach
Spleen
Kidney
Ovary
Ureter
Left horn uterus
Descending colon
Rectum
Vagina
Urethra
Urinary bladder
Body of uterus
Jejunum
Spleen
Liver
Caudal lobe of lung
Heart
Caudal part of cranial lobe lung
Cranial part of cranial lobe lung

# 腹腔臟器觸診

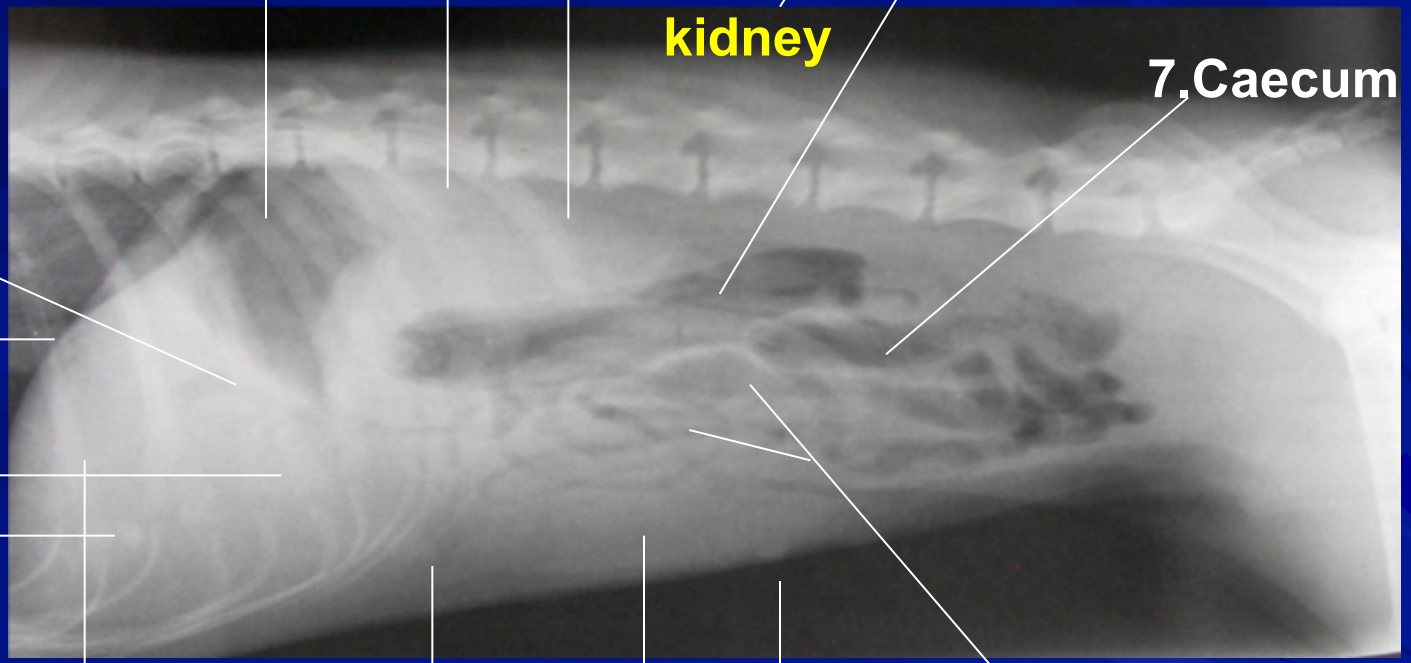
- 肝臟
  - 無法觸及，僅能到摸到肝臟尾端
- 脾臟
  - 腫大的脾會向腹側與尾側移位，可被辨識
- 腸道
  - 小腸：大小、厚度、內容物、疼痛反應
  - 結腸：寬度多變的堅硬管腔

    老貓便祕->極度擴張的結腸與極度堅硬

# 腸道聽診

- 腸鳴
  - 蠕動 + 液體 + 氣體
- 濺潑聲
  - 密閉空間內含大量氣體與大量液體
  - 上腹濺潑聲：胃排空不良
  - 全腹濺潑聲：腸道局部阻塞或蠕動全面喪失

# Left lateral recumbent projection



2.5-3X length of the 2nd lumbar body

3.Gas in gastric fundus

9.Dorsal extremity of spleen

8.Colon (ascending and descending)

11.Left kidney

7.Caecum

5.Dueodeum

1.Diaphragm

4.Pyloric part

Nipple

2.Liver

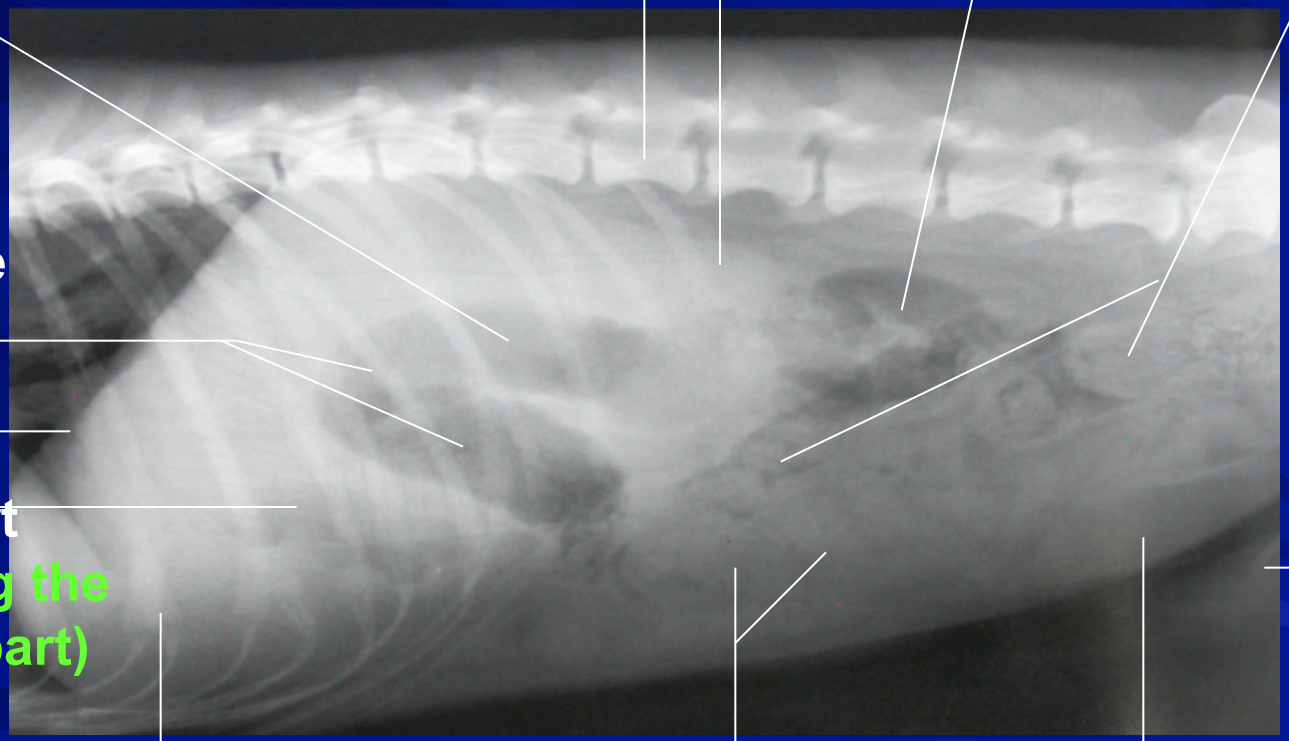
2'.Caudoventral margin of liver

Nipple

10.Spleen

6.Jejunum and ileum

# Left lateral recumbent, cranially centred



5.Ascending colon

9.Right kidney

2nd lumbar vertebra

4.Caecum

7.Descending colon

6.Transverse colon

1.Diaphragm

3.Pyloric part (gas entering the uppermost part)

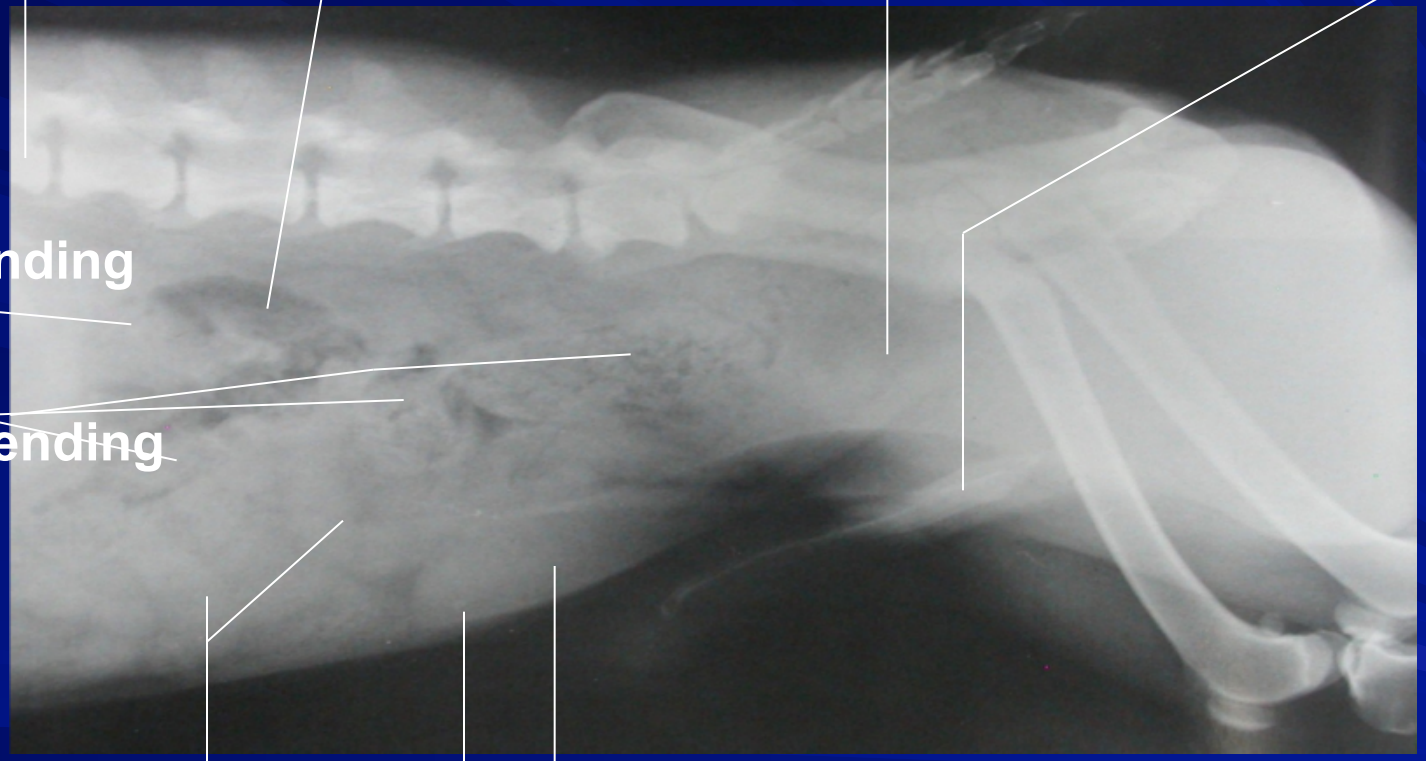
Os penis

2.Liver

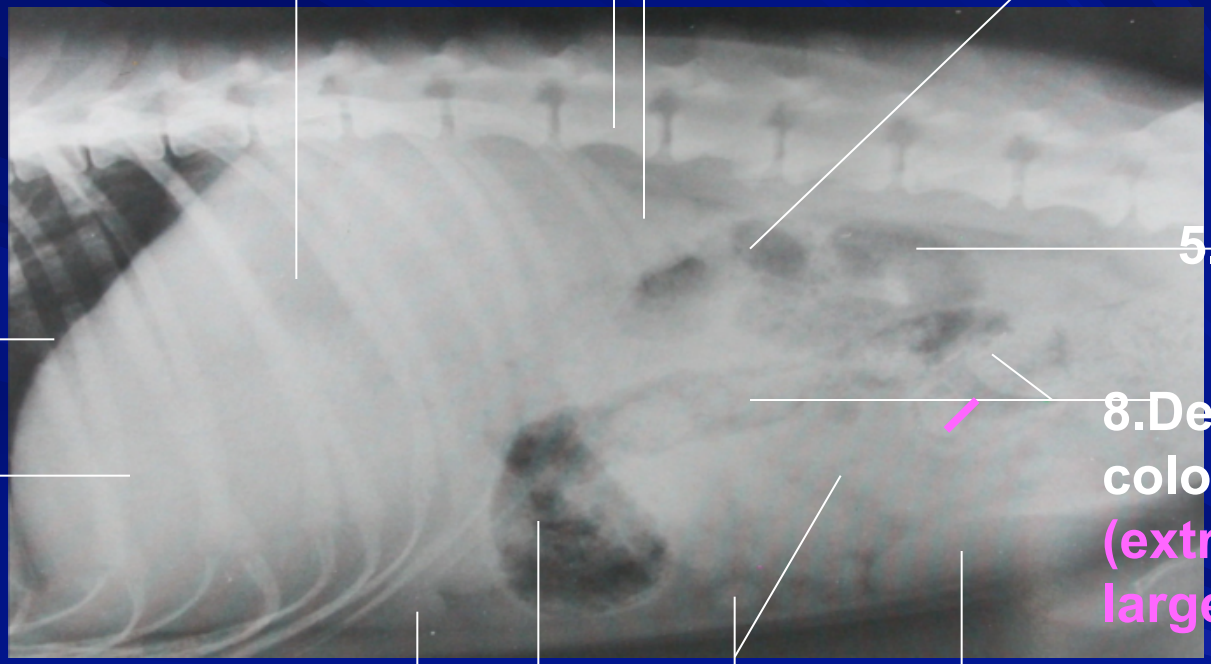
3.Jejunum and ileum

8.Ventral extremity of spleen

# Left lateral recumbent, caudally centred



# Right lateral recumbent, cranially centred



3.Gas in gastric fundus

2nd lumbar vertebra

6.Ascending colon

10.Right kidney

5.Caecum

1.Diaphragm

2.Liver

8.Descending colon (extra bend, large breed)

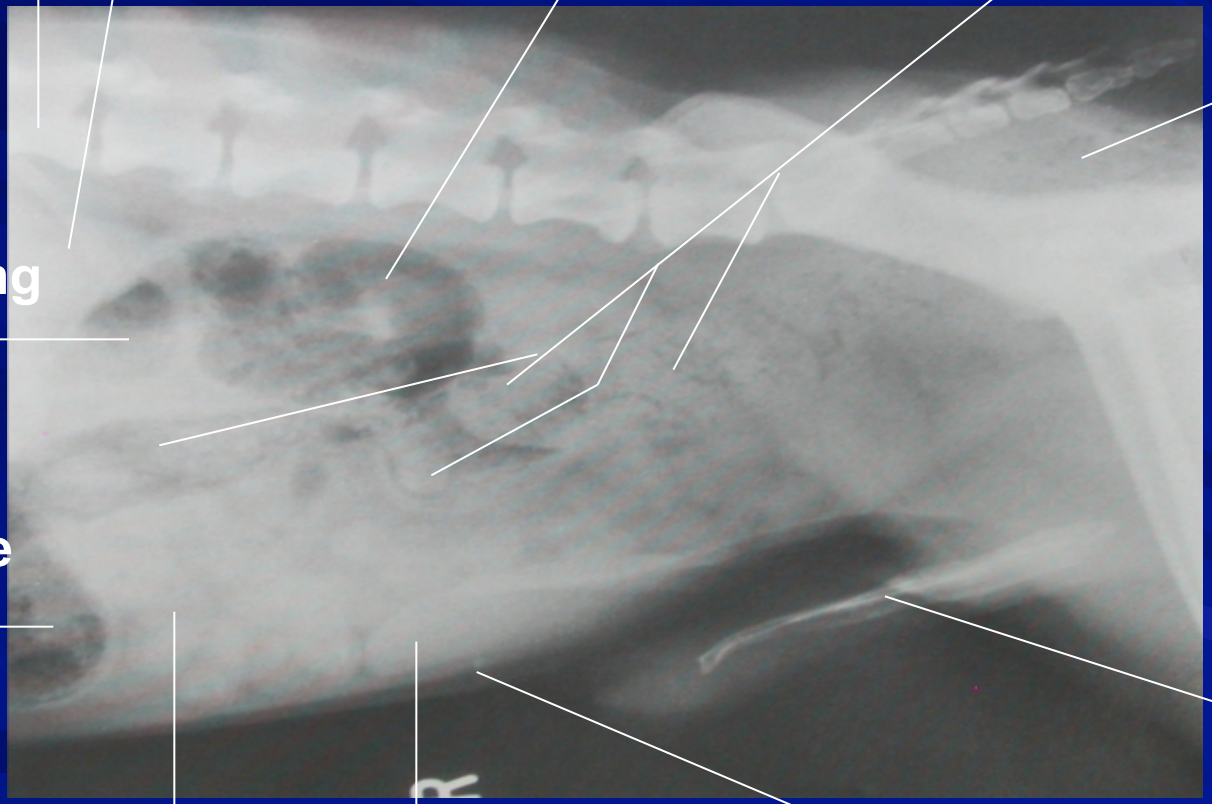
2'.Caudoventral margin of liver

7.Transverse colon

4.Jejunum and ileum

9.Ventral extremity of Spleen

# Right lateral recumbent, caudally centred



# Ventrodorsal projection



1.Diaphragm

4.Duodenum

3.Ingesta within gastric lumen

2nd lumbar vetebra

6.Caecum

5.Jejunum and ileum

Nipple

2'.Lesser curvature

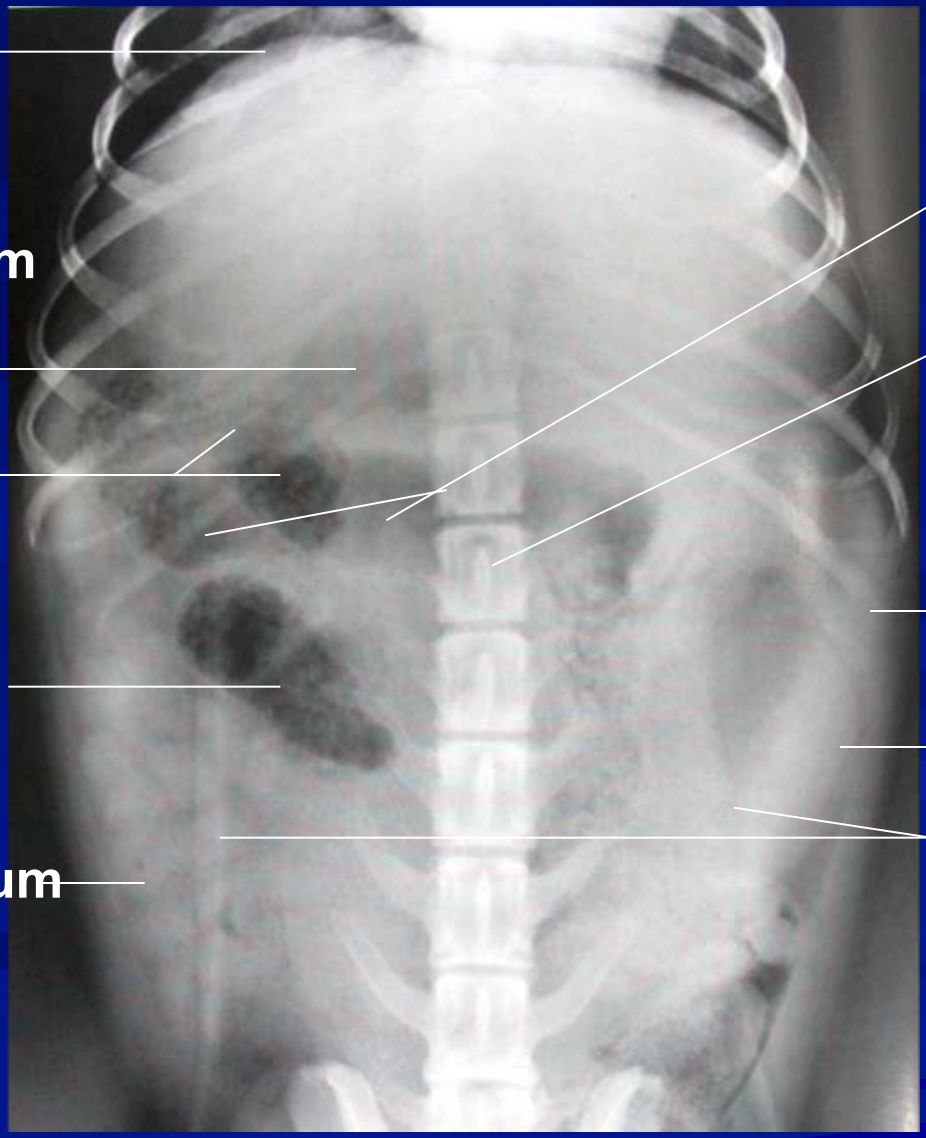
2.Gastric body

2".Greater curvature

8.Left kidney

7.Descending colon

# Ventrodorsal projection, cranially centred



1.Diaphragm

2.Pyloric antrum

5.Ascending colon

4.Caecum

3.Jejunum and ileum

6.Transverse colon

2nd lumbar vetebra

8.Dorsal extremity of spleen

7.Left kidney

Skin fold

# Ventrodorsal projection, caudally centred



3.Ascending colon

2.Caecum

Skin fold

1.Jejunum and ileum

Preputial shadow

2nd lumbar vetebra

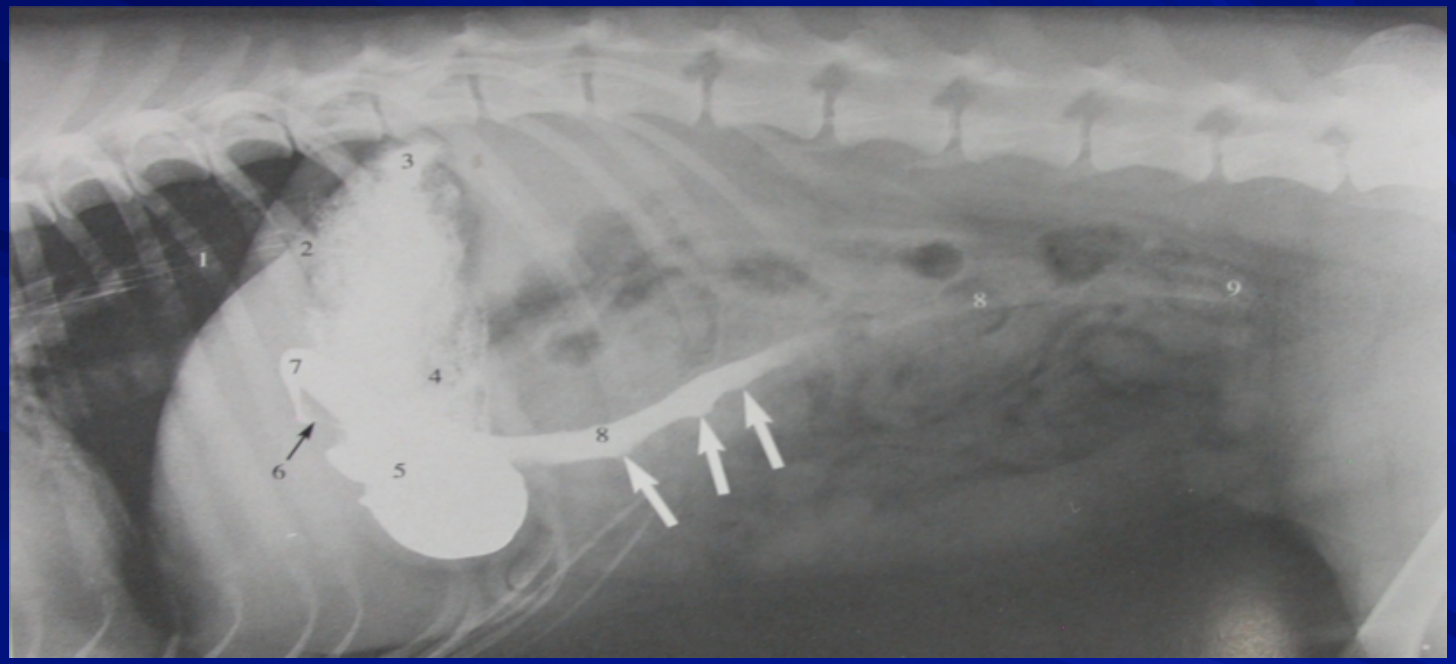
6.Left kidney

4.Descending colon

5.Rectum

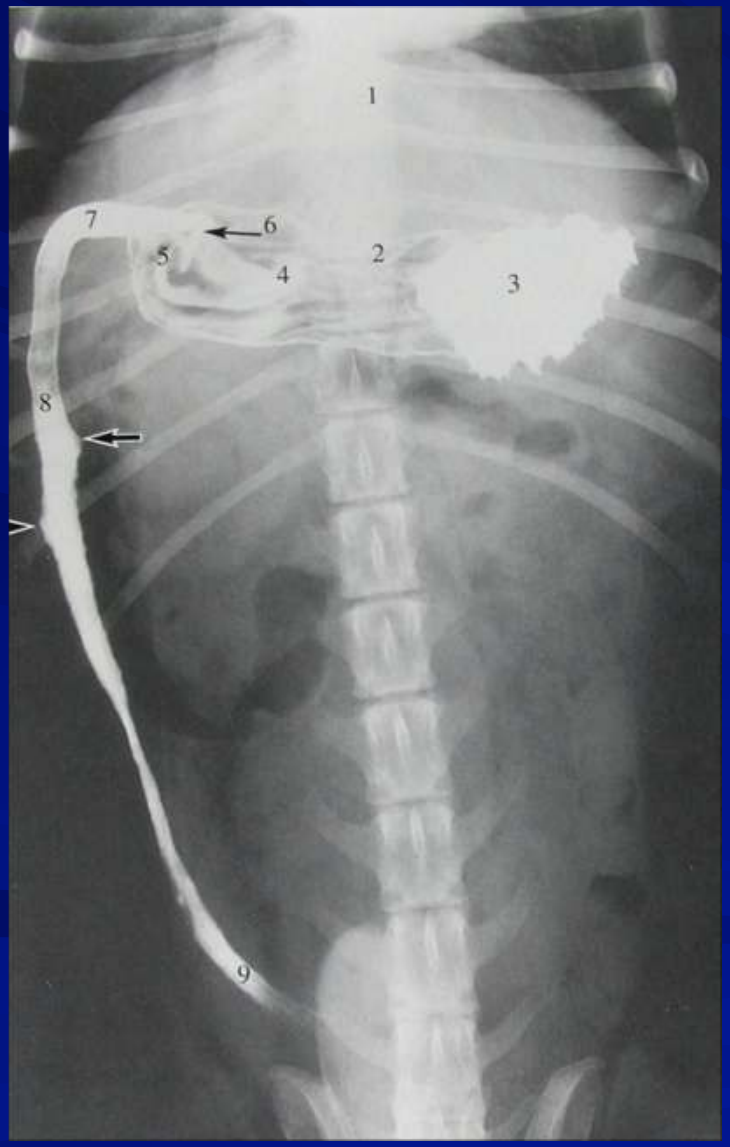
# Immediately after barium administration

## Right lateral recumbent



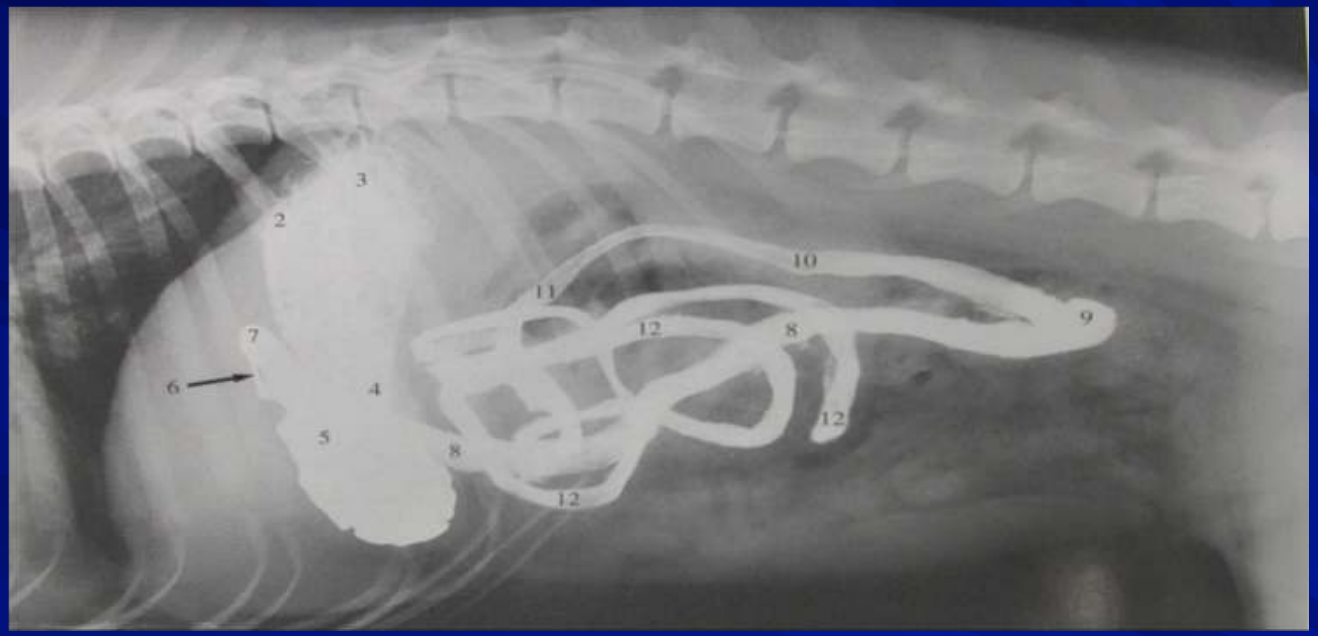

1. Oesophagus
2. Cardia of the stomach
3. Fundus of the stomach
4. Body of the stomach
5. Pyloric antrum
6. Pyloric canal
7. Cranial flexure of duodenum
8. Descending portion of duodenum
9. Caudal flexure of duodenum

# Immediately after barium administration




1. Oesophagus
2. Cardia of the stomach
3. Fundus of the stomach
4. Body of the stomach
5. Pyloric antrum
6. Pyloric canal
7. Cranial flexure of duodenum
8. Descending portion of duodenum
9. Caudal flexure of duodenum

# 30min after barium administration



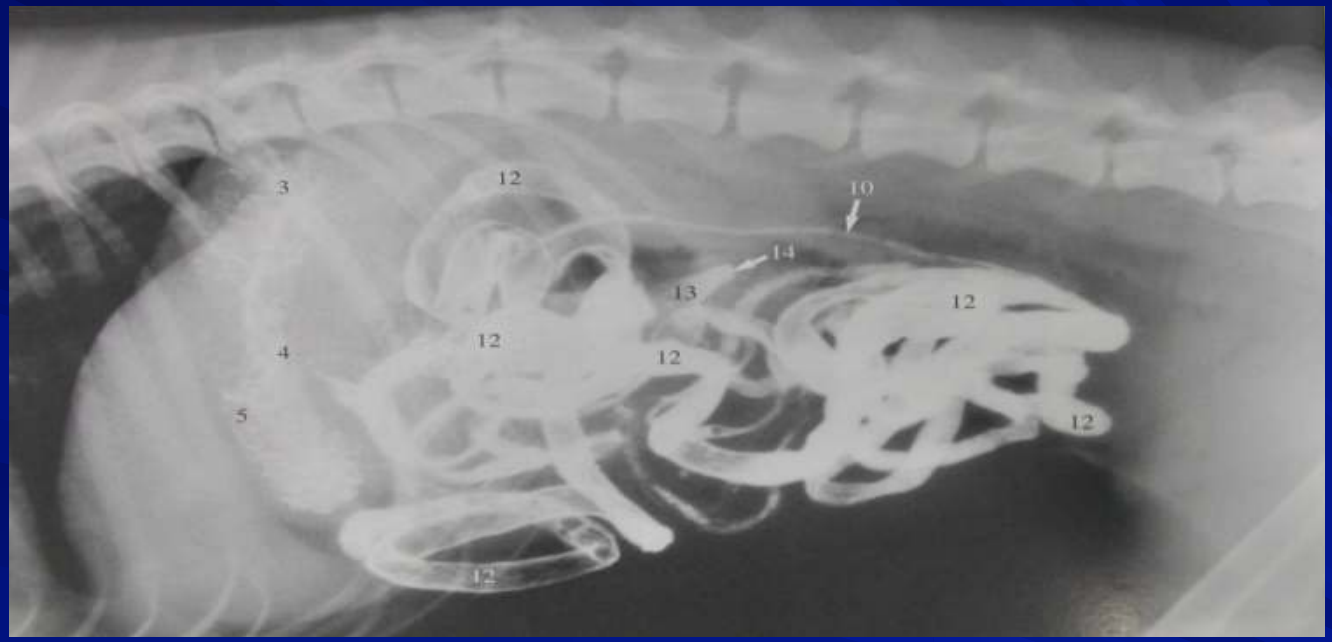

2. Cardia of the stomach
3. Fundus of the stomach
4. Body of the stomach
5. Pyloric antrum
6. Pyloric canal
7. Cranial flexure of duodenum
8. Descending portion of duodenum
9. Caudal flexure of duodenum
10. Ascending portion of duodenum
11. Dudenojejunal flexure
12. Loops of jejunum and ileum

# 30 min after barium administration




3. Fundus of the stomach
4. Body of the stomach
5. Pyloric antrum
7. Cranial flexure of duodenum
8. Descending portion of duodenum
9. Caudal flexure of duodenum
10. Ascending portion of duodenum
11. Dudenojejunal flexure
12. Loops of jejunum and ileum

# 1 hr after barium administration



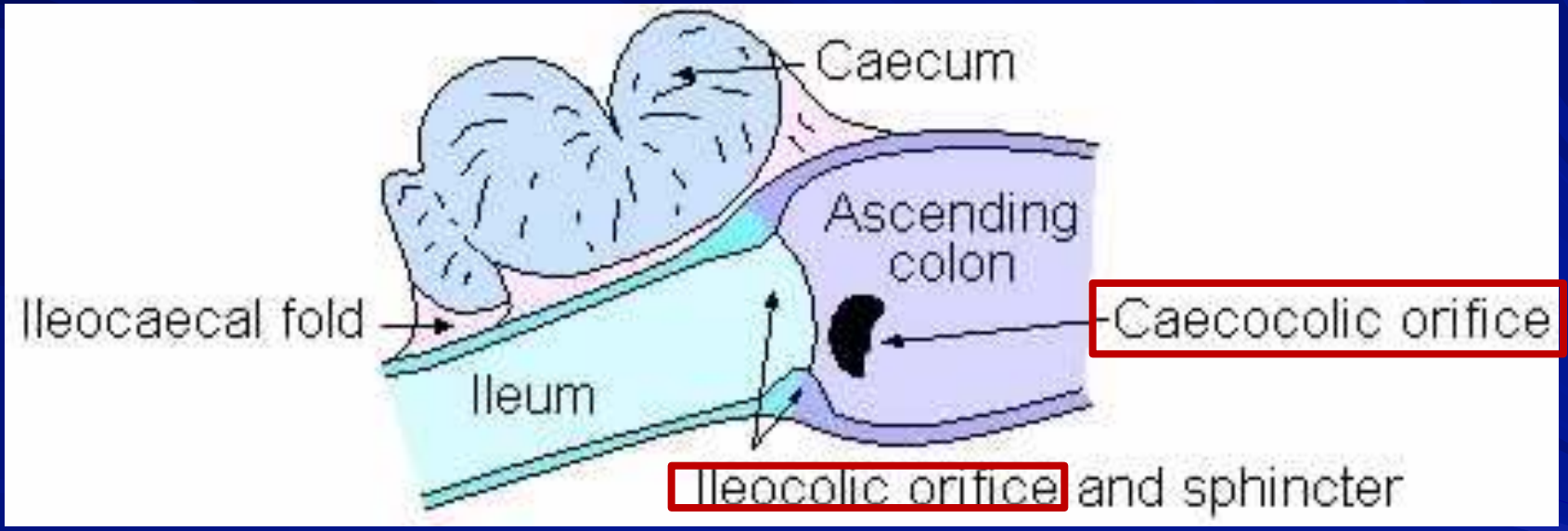

3. Fundus of the stomach
4. Body of the stomach
5. Pyloric antrum
10. Ascending portion of duodenum
12. Loops of jejunum and ileum
13. Terminal portion of ileum
14. Ileocolic junction

# 1 hr after barium administration

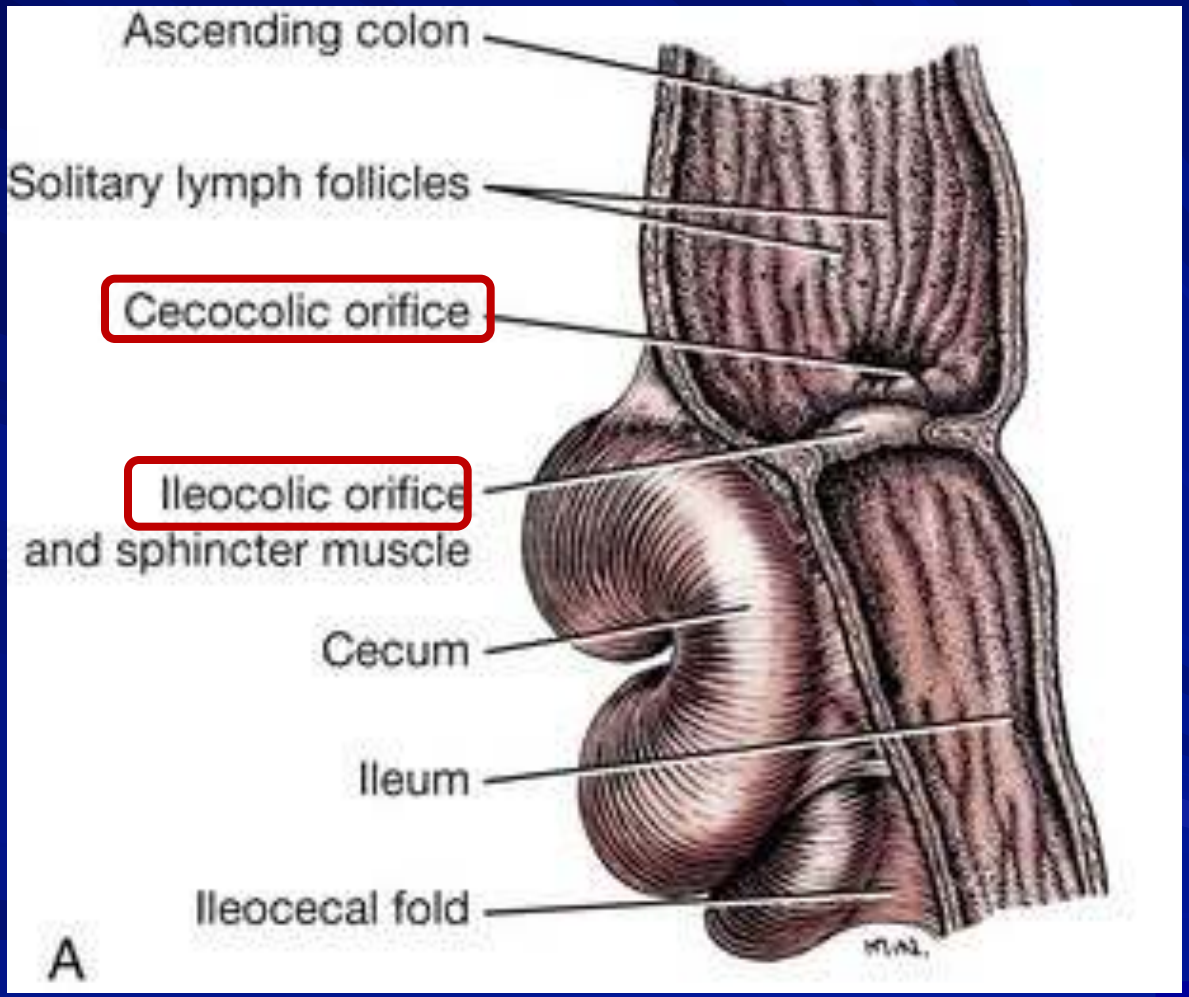



3. Fundus of the stomach
4. Body of the stomach
7. Cranial flexure of duodenum
8. Descending portion of duodenum
10. Ascending portion of duodenum
12. Loops of jejunum and ileum
13. Terminal portion of ileum
14. Ileocolic junction

# Dog caecum

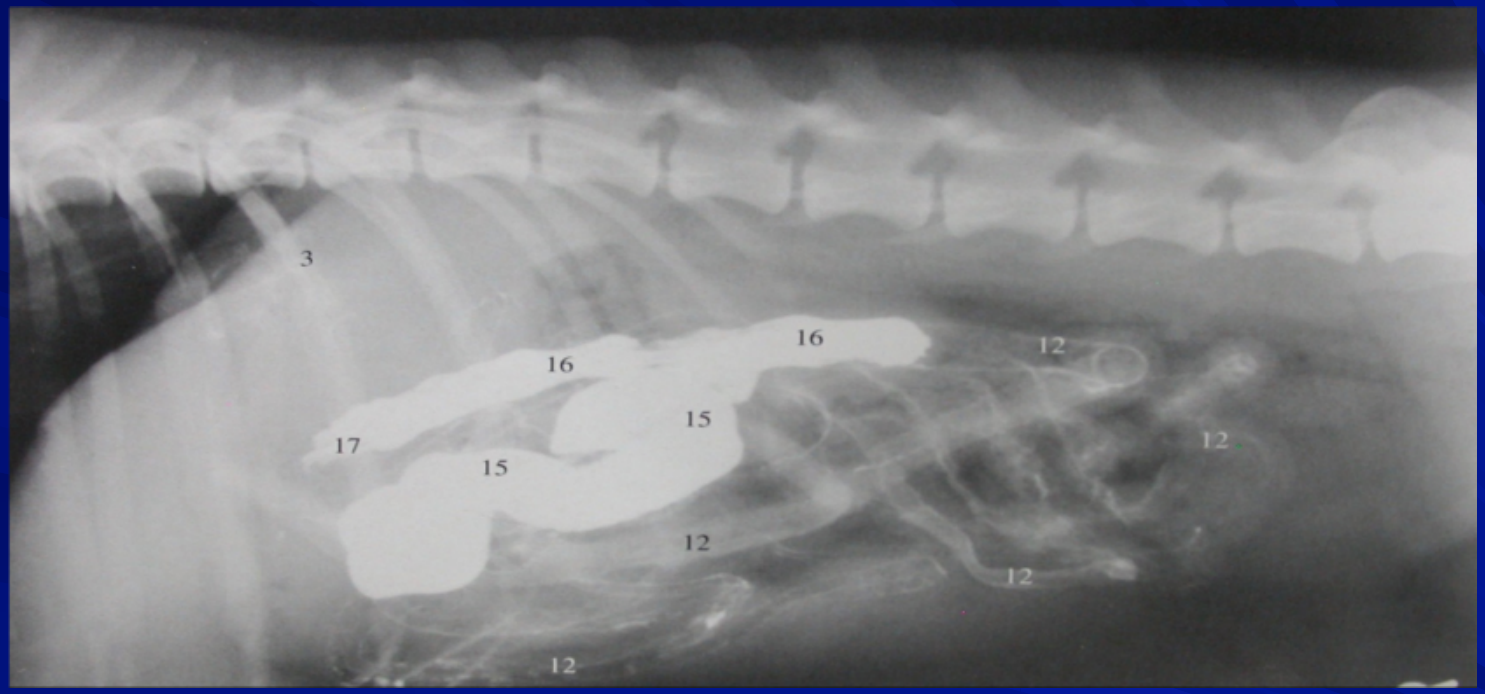

Caecum
Ascending colon
Ileocaecal fold
Caecocolic orifice
Ileum
Ileocolic orifice and sphincter

# Canine ileocolic junction




https://veteriankey.com/colon/

# 4 hr after barium administration



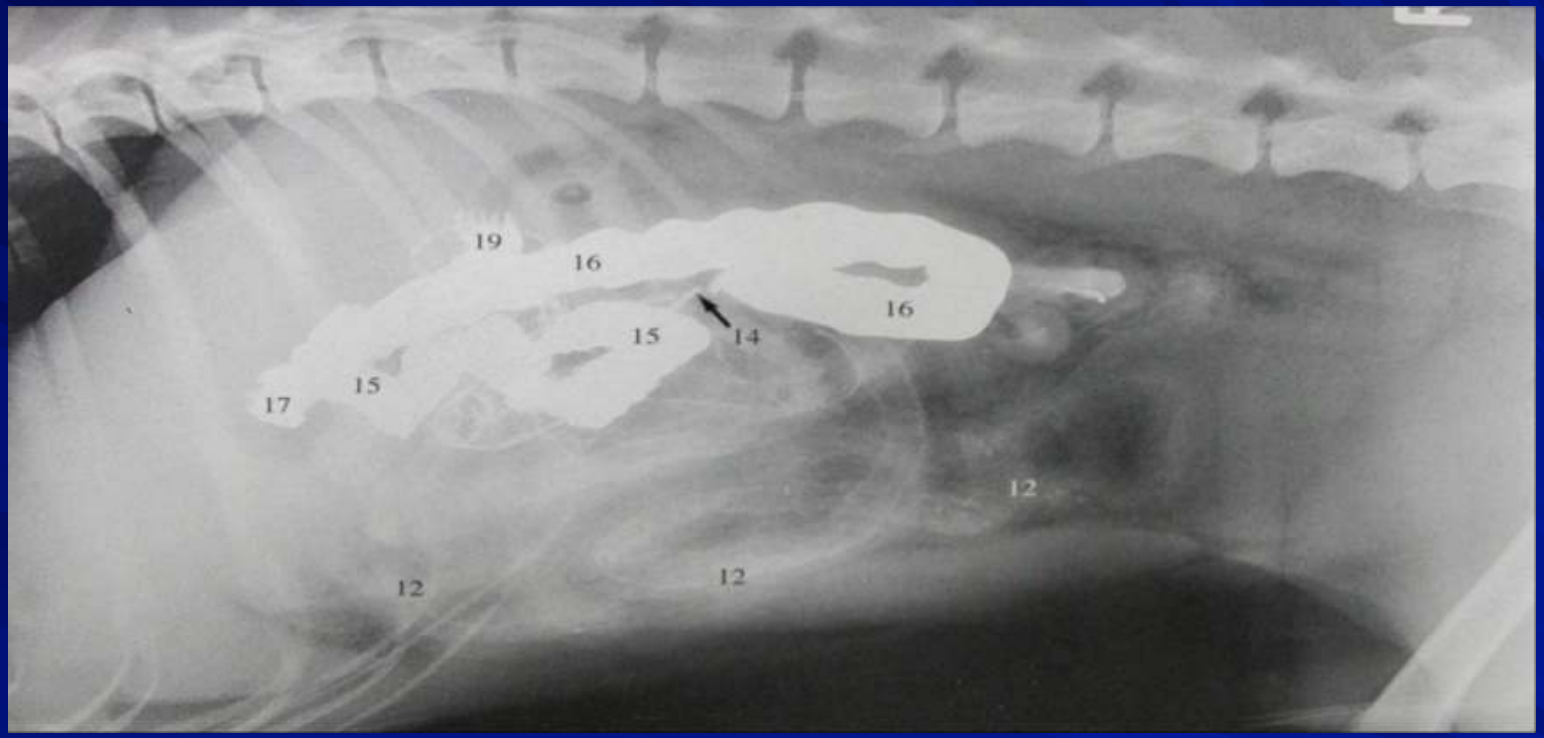

3. Fundus of the stomach
12. Loops of jejunum and ileum
15. Caecum
16. Ascending portion of colon
17. Right colic flexure or hepatic flexure of colon

# 4 hr after barium administration




3. Fundus of the stomach
7. Cranial flexure of duodenum
8. Descending portion of duodenum
9. Caudal flexure of duodenum
10. Ascending portion of duodenum
12 Loops of jejunum and ileum
13. Terminal protion of ileum
14. Ileocolic junction
15. Caecum
16. Ascending portion of colon
17. Right colic flexure or hepatic flexure of colon
18. Transverse portion of colon

# 6 hr after barium administration




12. Loops of jejunum and ileum
14. Ileocolic junction
15. Caecum
16. Ascending portion of colon
17. Right colic flexure or hepatic flexure of colon
19. Left colic flexure or splenic flexure of colon

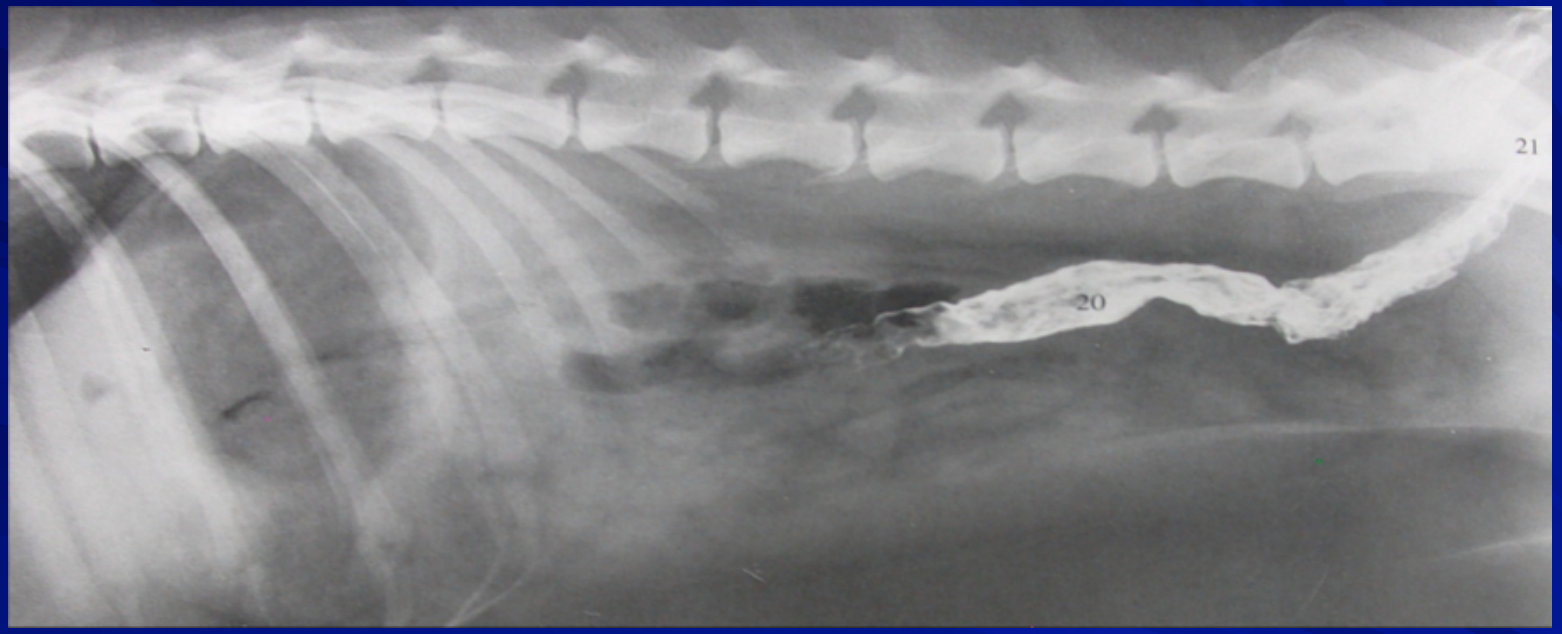
# 6 hr after barium administration




7. Cranial flexure of duodenum
8. Descending portion of duodenum
9. Caudal flexure of duodenum
10. Ascending portion of duodenum

12 Loops of jejunum and ileum
13. Terminal protion of ileum
14. Ileocolic junction
15. Caecum
16. Ascending portion of colon
17. Right colic flexure or hepatic flexure of colon
18. Transverse portion of colon
19. Left colic flexure or splenic flexure of colon

# 12 hr after barium administration



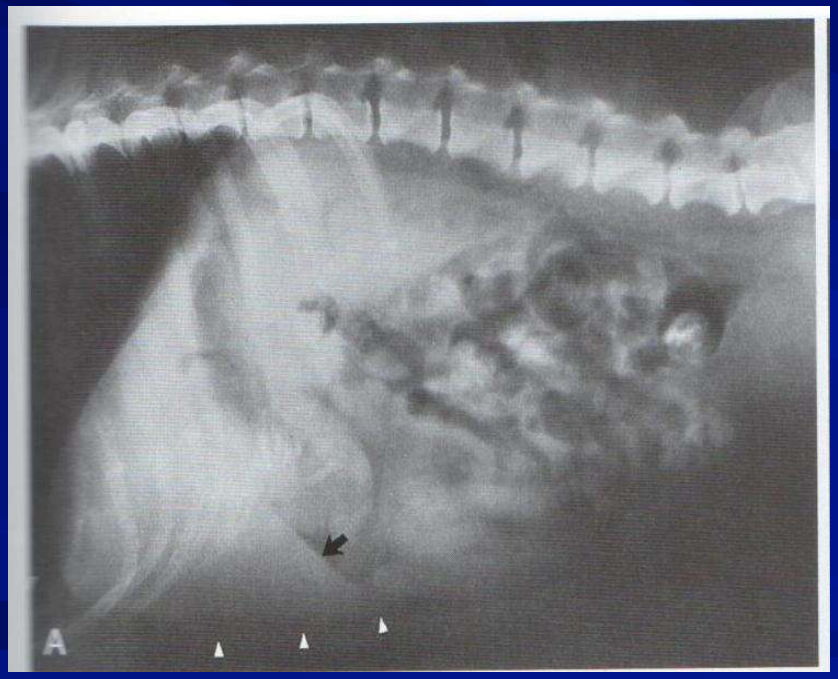

20. Descending portion of colon
21. Rectum

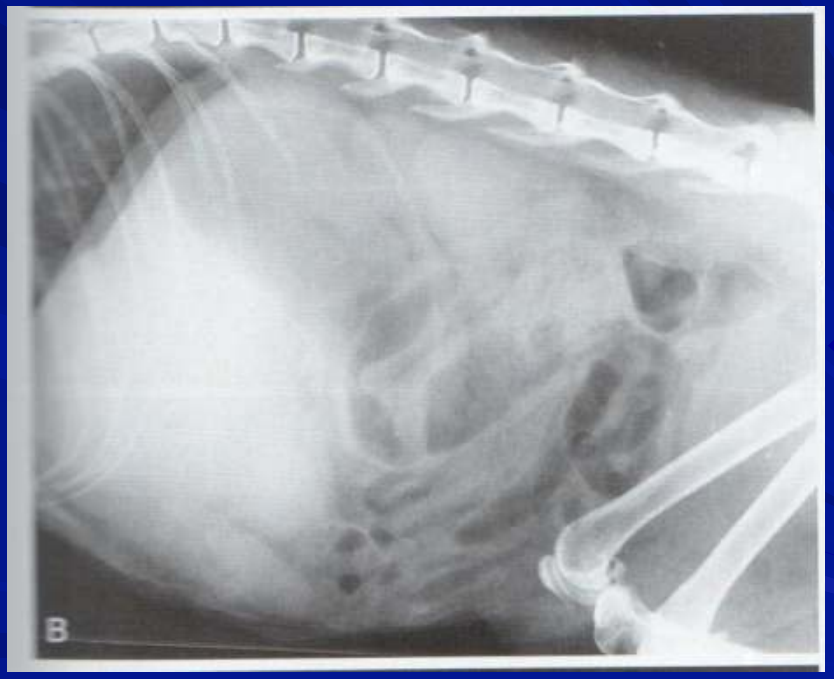
# 12 hr after barium administration




20. Descending portion of colon
21. Rectum

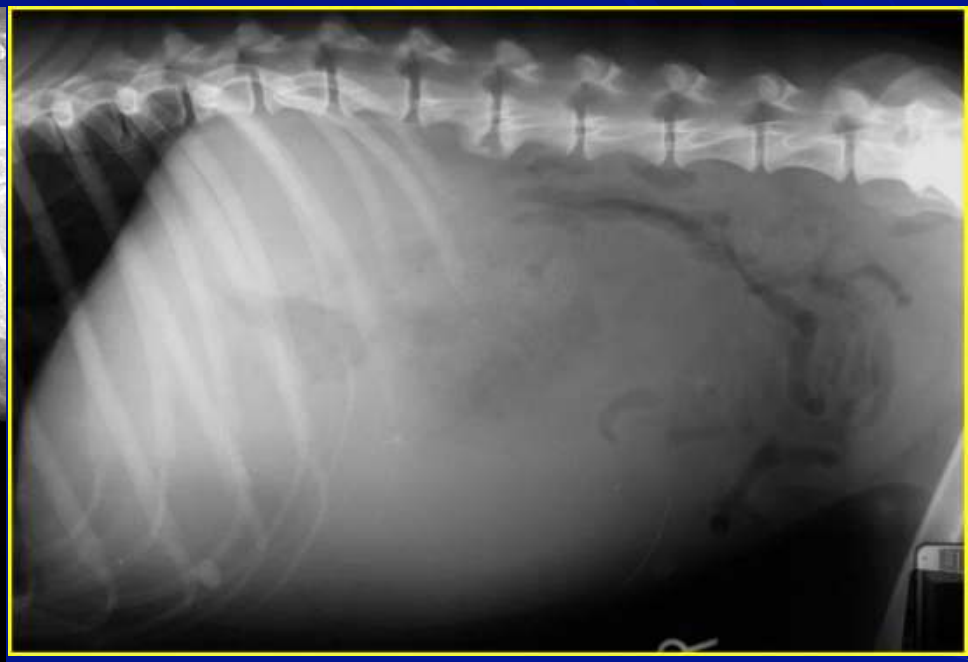
# Enlarged liver



# Enlarged spleen



# Foreign body in the stomach

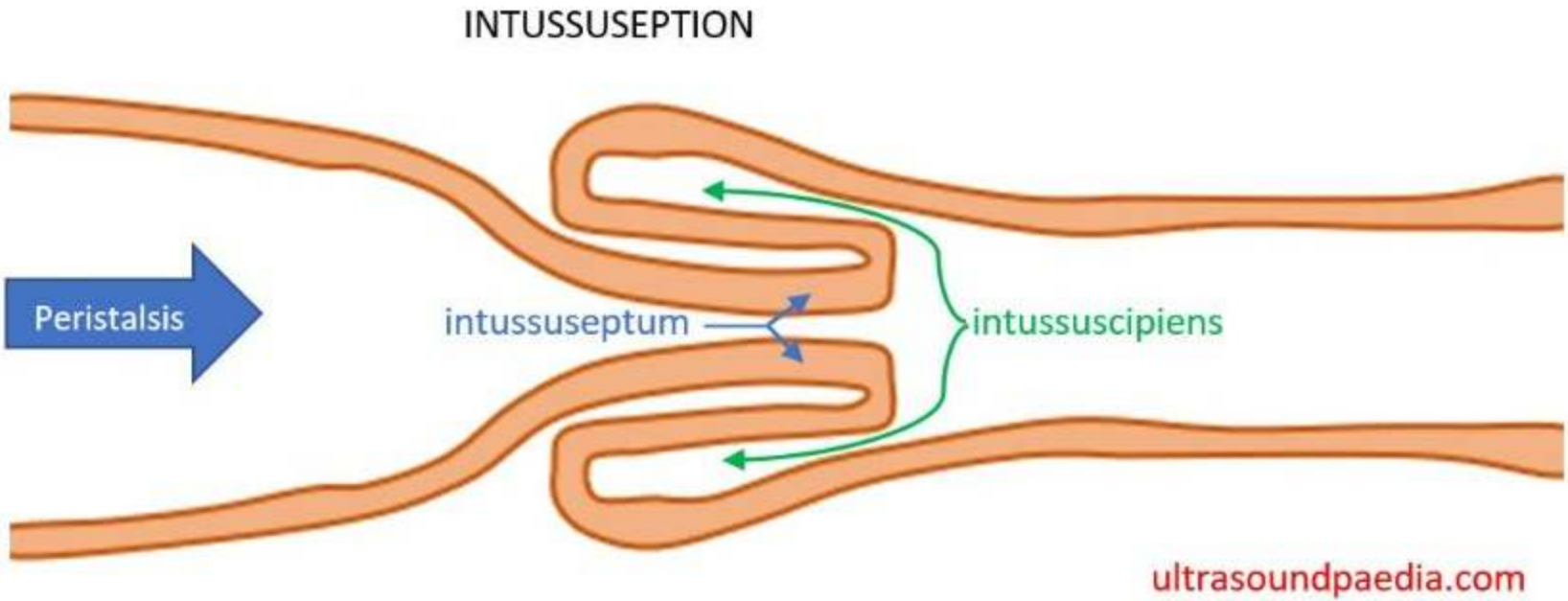


Plain view



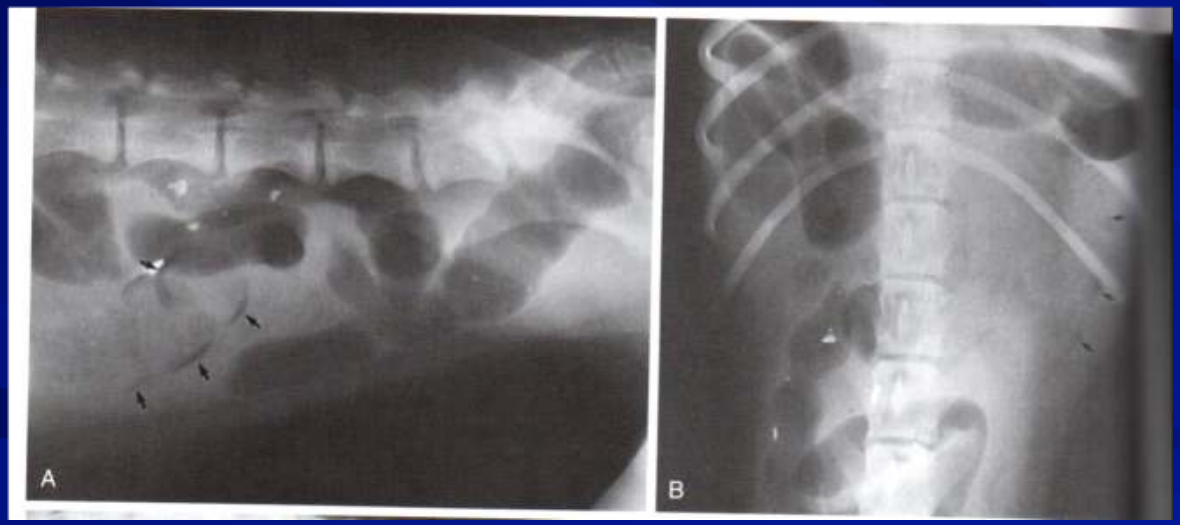

After barium administration

# Intussusception




https://ultrasoundpaedia.com/intussusception/

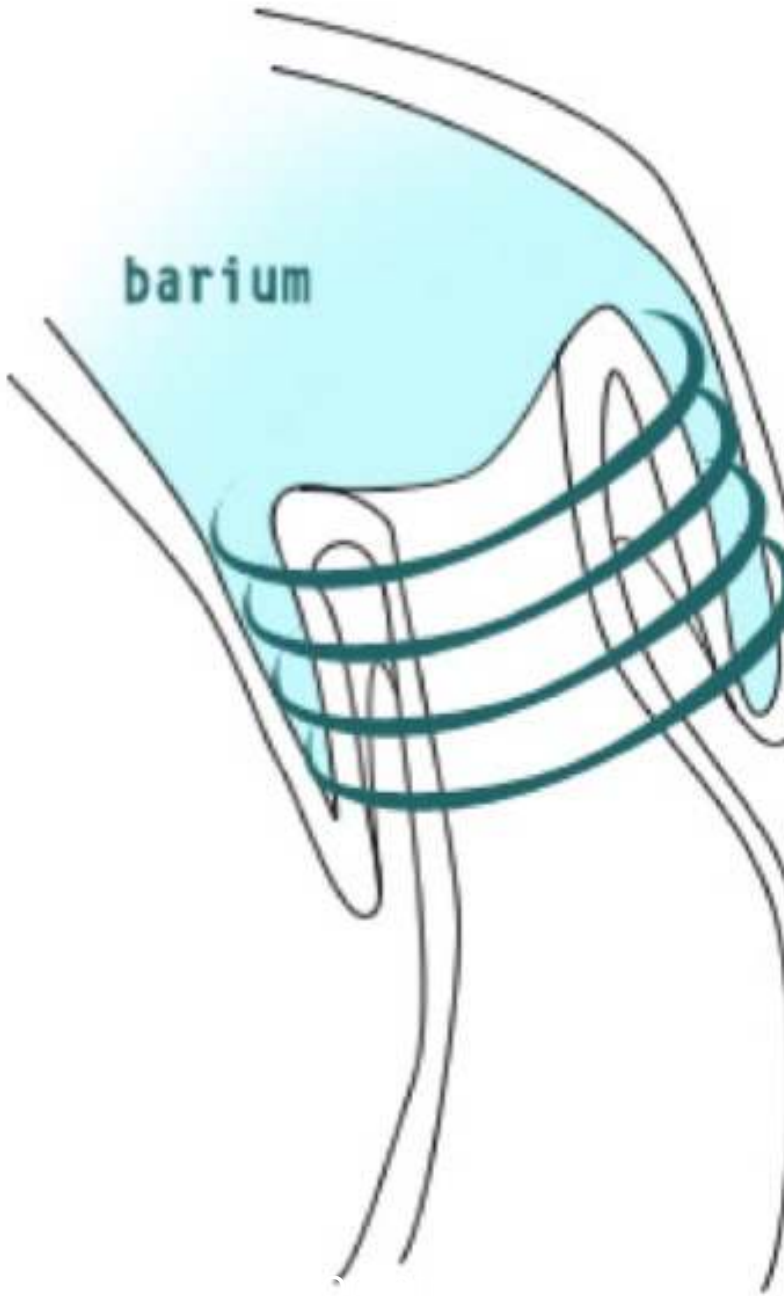
# Intussusception
# -plain radiographs





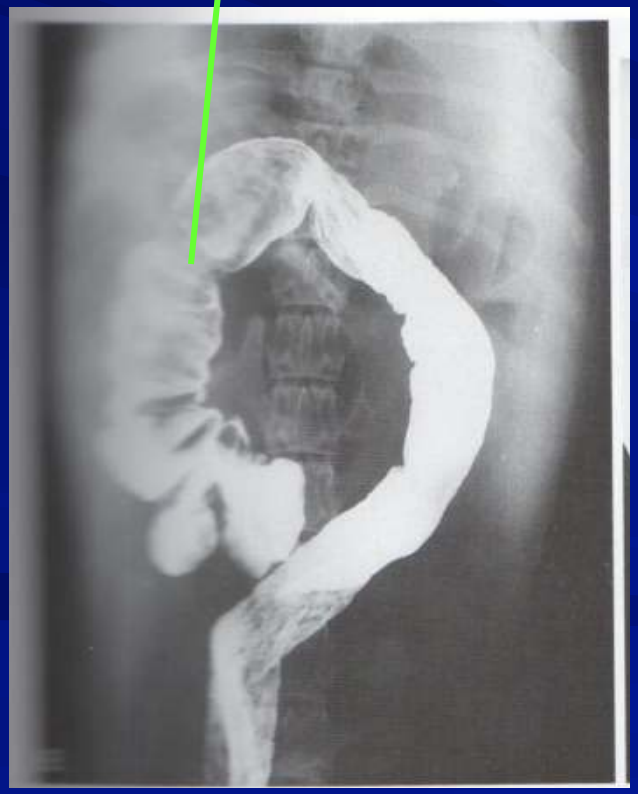







http://www2.cmu.edu.tw/~cmcmd/ctanatomy/cli

# Intussusception-barium enema






# 臨床暨分子醫學研究室
# 誠徵研究生

- 臨床醫學與分子診斷學的結合性研究
- 使用創新的分子診斷技術快速診斷犬神經行為病原性的疾病、犬與貓的呼吸道病原性的疾病
- 結合病史與臨床症狀，作流行病學的統計分析
- 參住院醫師與的實習
- **提供研究生獎助學金** (於非住院醫師期間)
- 歡迎欲參加研究生甄試的同學與老師聯絡：
  702ntu@gmail.com

  **02 33661299**

# Discussion

1. Please identify each body part on this radiograph.
2. What have you seen on this radiograph? Give a diagnosis and explain.